Stat 437 Study Guide

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Study Guide: Topics Maybe Covered on the Exam

General Expectations

You should be able to:

- Perform calculations and clearly show intermediate steps
- Interpret results in clinical, public health, or policy context
- Identify limitations, biases, and assumptions
- Propose appropriate study designs and analytic strategies
- Connect statistical methods to causal questions

The exam will emphasize **applied reasoning**, not memorization.

1. Disease Occurrence: Morbidity and Mortality

You should be able to:

Definitions and Concepts

- Distinguish between incidence (cumulative vs rate), prevalence (point vs period), and mortality
- Identify numerator, denominator, and time window for each measure

Calculations

- Compute cumulative incidence
- Compute incidence rate using person-time
- Compute point and period prevalence
- Compute mortality rates

Interpretation

- Interpret measures in context

- Compare incidence vs prevalence

Reasoning

- Understand relationship between incidence, duration, and prevalence
- Explain how detection, treatment, and survival affect measures

Study Design Connections

- Propose designs to study disease risk
- Identify appropriate populations and follow-up

2. Risk, 2×2 Tables, and Measures of Association

You should be able to:

Calculations

- Construct and interpret 2×2 tables
- Compute risk, risk difference (RD), relative risk (RR), and odds ratio (OR)

Interpretation

- Interpret RD, RR, and OR in context
- Distinguish absolute vs relative measures
- Recognize when OR approximates RR

Statistical Inference

- State hypotheses (e.g., $H_0 : p_1 = p_0$, $RR = 1$, $OR = 1$)
- Describe tests (two-sample proportion, chi-square)
- Understand pooled vs unpooled variance
- Interpret p-values and confidence intervals

Conceptual Understanding

- Understand equivalence of tests in 2×2 settings
- Distinguish statistical vs clinical significance

Bias and Causal Interpretation

- Recognize confounding in observational studies

3. Study Design, Randomized Trials, and Power

You should be able to:

Study Design Basics

- Distinguish randomized controlled trials (RCTs), cohort studies, and case-control studies
- Explain the strengths and limitations of each design
- Identify when randomization is feasible or not feasible
- Explain why randomization helps with confounding

Causal Interpretation

- Explain why association does not imply causation
- Recognize confounding by indication in observational studies
- Distinguish internal validity from external validity
- Identify how selection bias, measurement bias, and reverse causation can distort conclusions

Power and Sample Size

- Explain the meaning of statistical power
- State what Type I error and Type II error mean
- Interpret α, power, and effect size in the context of a study
- Recognize how larger sample size, larger effect size, and less variability affect power
- Understand why smaller effects require larger sample sizes to detect
- Explain the role of two-sided vs one-sided tests in planning

Analytic Planning

- Determine whether a study is trying to test a difference in means, proportions, risks, or time-to-event outcomes
- Identify the appropriate endpoint for a trial or observational study
- Choose a summary measure that matches the outcome (e.g., risk difference, relative risk, hazard ratio)
- Explain how missing data, unequal follow-up, or censoring affect study planning

Simulation and Robustness

- Understand why simulation may be used to assess power when analytic formulas are not adequate
- Explain how non-normality, heavy tails, or unequal variances can affect power
- Recognize why simulated power may differ from analytic power

- Interpret simulation output in the context of trial planning

Randomized Trial Concepts

- Understand the role of randomization in balancing measured and unmeasured confounders
- Explain why blinding and allocation concealment matter
- Understand intention-to-treat as a principle for analyzing randomized trials
- Distinguish treatment effect from adherence or per-protocol effects

Design Questions You May Need to Answer

- What is the target population?
- What is the intervention or exposure?
- What is the control or comparison group?
- What is the primary endpoint?
- What covariates should be collected at baseline?
- What sample size would be needed to have adequate power?
- What assumptions are needed for the analysis to be valid?

4. Diagnostic and Screening Tests

You should be able to:

Definitions

- Sensitivity, specificity
- Positive and negative predictive values (PPV, NPV)

Calculations

- Compute test characteristics from 2×2 tables
- Apply Bayes' theorem to compute PPV or NPV

Derivations

- Derive PPV or NPV using Bayes' rule and total probability

Interpretation

- Interpret predictive values in context
- Explain dependence on prevalence

Sequential and Parallel Testing

- Derive net sensitivity and specificity

- Understand tradeoffs between sensitivity and specificity

Decision-Making

- Choose testing strategies based on clinical context
- Evaluate consequences of false positives and false negatives

5. Causal Inference and DAGs

You should be able to:

DAG Construction

- Draw DAGs from a scenario
- Justify relationships between variables

Identify Variable Roles

- Confounders
- Mediators
- Colliders

Adjustment Decisions

- Identify which variables to adjust for
- Avoid adjusting for mediators and colliders

Backdoor Paths

- Identify and block backdoor paths

Bias Mechanisms

- Confounding bias
- Collider bias
- Selection bias

Propensity Score Methods

- Understand goal and limitations

6. Binary Outcomes and Logistic Regression

You should be able to:

Foundations

- Understand binary outcomes and odds
- Define log-odds (logit)

Model Specification

- Write logistic regression models

Interpretation

- Interpret $\exp(\beta)$ as odds ratios
- Interpret effects for continuous and categorical predictors

Confounding Adjustment

- Understand role of covariates
- Recognize omitted variable bias

Model Building

- Assess nonlinearity
- Understand interactions

Model Evaluation

- Compare models using AIC
- Understand fit vs interpretability tradeoffs

7. Endpoint Selection and Model Choice

You should be able to:

Match Outcomes to Models

- Binary $\rightarrow$ logistic regression
- Ordinal $\rightarrow$ cumulative logit model
- Count $\rightarrow$ Poisson / negative binomial
- Time-to-event $\rightarrow$ survival analysis

Interpretation

- Distinguish odds ratios, rate ratios, and hazard ratios

Conceptual Differences

- Risk vs rate vs hazard
- Timing vs binary outcomes

Decision-Making

- Choose appropriate outcome and model
- Understand tradeoffs between simplicity and information

8. Survival Analysis

You should be able to:

Core Concepts

- Survival function $S(t)$
- Hazard function $h(t)$
- Relationship between survival and hazard

Kaplan–Meier Estimation

- Compute KM estimates by hand
- Track risk sets, events, and censoring
- Interpret survival curves

Censoring

- Understand right censoring
- Explain non-informative censoring
- Recognize informative censoring

Comparing Groups

- Understand log-rank test and its assumptions

Cox Proportional Hazards Model

- Interpret hazard ratios
- Understand proportional hazards assumption

Study Design and Bias

- Choose appropriate endpoints
- Understand impact of censoring

Final Note: You are expected to apply these concepts to realistic scenarios, justify your reasoning, and clearly interpret results in context.